$P(X<1) = 0$ for $-\infty < x < 1$.

$$F(1) = P(X \le 1) = \sum_{x_i \le x} P(X = x_i) = \sum_{-\infty}^{1} P(X = x) = P(X<1) + P(X=1) = 0 + \frac{1}{12} = \frac{1}{12}.$$

$$F(2) = P(X \le 2) = \sum_{-\infty}^{2} P(X = x) = P(X<1) + P(X=1) + P(X=2).$$

$$= 0 + \frac{1}{12} + \frac{5}{12} = \frac{1}{2}.$$

$$F(3) = P(X \le 3) = \sum_{-\infty}^{3} P(X = x) = P(X<1) + P(X=1) + P(X=2) + P(X=3).$$

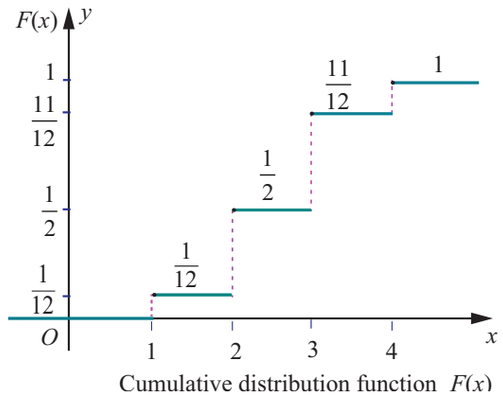
$$= 0 + \frac{1}{12} + \frac{5}{12} + \frac{5}{12} = \frac{11}{12}.$$

$$F(4) = P(X \le 4) = \sum_{-\infty}^{4} P(X = x) = P(X<1) + P(X=1) + P(X=2) + P(X=3) + P(X=4).$$

$$= 0 + \frac{1}{12} + \frac{5}{12} + \frac{5}{12} + \frac{1}{12} = 1.$$

Therefore the cumulative distribution function is

$$F(x) = \begin{cases} 0, & -\infty < x < 1 \\ \frac{1}{12}, & 1 \le x < 2 \\ \frac{1}{2}, & 2 \le x < 3 \\ \frac{11}{12}, & 3 \le x < 4 \\ 1, & 4 \le x < \infty \end{cases}$$



Cumulative distribution function $F(x)$

Fig. 11.6

(ii) $P(X \le 3) = F(3) = \frac{11}{12}$.

(iii) $P(X \ge 2) = 1 - P(X < 2) = 1 - P(X \le 1) = 1 - F(1) = 1 - \frac{1}{12} = \frac{11}{12}$.

Example 11.8

A six sided die is marked '1' on one face, '2' on two of its faces, and '3' on remaining three faces. The die is rolled twice. If X denotes the total score in two throws.



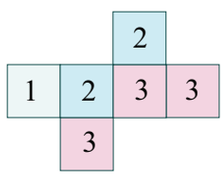


Fig. 11.7

(i) Find the probability mass function.

(ii) Find the cumulative distribution function.

(iii) Find $P(3 \le X < 6)$ (iv) Find $P(X \ge 4)$.

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